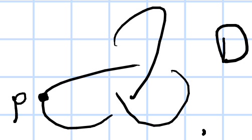
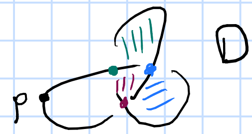
Some invariants of knots

Let $K \subseteq S^3$ be a knot, D its diagram with a marked point $p \in D$, eg

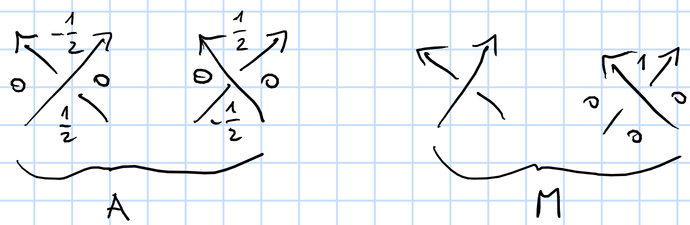


Def: A Kauffman state $\mathcal{K}$ is a bijection between the set of crossings of D and the set of domains not containing p, s.t. $c \in \overline{\mathcal{K}(c)}$ for every crossing c.

eg



Let



and $A(\mathcal{K}) = \sum_{\text{crossings}} A(\mathcal{K}(c))$, $M(\mathcal{K}) = \sum M(\mathcal{K}(c)) \in \mathbb{Z}$, and

$$E_{(D,p)}(q,t) = \sum q^{M(\mathcal{K})} t^{A(\mathcal{K})}.$$

This is <u>not</u> a knot invariant.

Def: Let
$$\Delta_K(t) = E_{(D,p)}(-1,t).$$

<u>Thm</u>: This is the Alexander polynomial.

Note: $\deg \Delta_K$ bounds the genus of the Seifert surface from below. If K if fibred, ie $S^3 \setminus K$ fibres over S^1, then its leading coefficient is ± 1.

Def: Let $\widehat{CFK}$ be the bigraded vector space over $\mathbb{Z}/2\mathbb{Z}$ generated by Kauffman states, $\widehat{CFK}^-$ the bigraded free